

Death/ERK parameter sweep and pooled Bonferroni analysis

To assess whether the spatial enrichment of observed cell deaths in the fixed T-shaped region is robust to death and ERK feedback parameters, we performed a death/ERK-only parameter sweep. The activation regime was kept fixed at

$$\lambda_{a,T} = 0.5, \quad \lambda_{a,c} = 0.005, \quad \beta_{a,R} = 2.5, \quad \beta_{a,T} = 3.0.$$

Only the death and ERK parameters were varied:

$$\lambda_d \in \{0.5, 1.0, 2.0\}, \quad \beta_{d,R} \in \{1.0, 2.0, 4.0\}, \quad \beta_{d,T} \in \{0.4, 0.8, 1.6\}.$$

This gives a full factorial design with

$$3 \times 3 \times 3 = 27$$

parameter settings. For each setting, five independent replicates were simulated. In each replicate, the first 500 observed deaths were used as burn-in and discarded from the statistical analysis. The next 1000 observed deaths were then analyzed.

Let $W = [0, L_x] \times [0, L_y]$ be the spatial domain, with $L_x = L_y = 10$. Let $T \subset W$ denote the fixed T-shaped region. Its geometric area fraction is

$$p_0 = \frac{|T|}{|W|} = 0.4444.$$

For each replicate, if k_T observed deaths fall inside T among n analyzed deaths, the observed T-zone fraction is

$$\hat{\pi}_T = \frac{k_T}{n}.$$

The T-zone density ratio is defined as

$$R_T = \frac{k_T/|T|}{(n - k_T)/(|W| - |T|)}.$$

Thus, $R_T > 1$ means that the density of observed deaths is higher inside the T-zone than outside it, while $R_T = 1$ corresponds to equal death density inside and outside the T-zone.

The statistical test for T-zone enrichment is the one-sided exact binomial test

$$H_0 : \pi_T = p_0, \quad H_A : \pi_T > p_0.$$

Under the null hypothesis,

$$K_T \sim \text{Binomial}(n, p_0),$$

and the one-sided p -value is

$$p_{\text{val}} = \mathbb{P}_{H_0}(K_T \geq k_T) = \sum_{j=k_T}^n \binom{n}{j} p_0^j (1 - p_0)^{n-j}.$$

For the pooled analysis, the five independent replicates of each parameter setting were combined. Since each replicate contributes 1000 analyzed observed deaths, each pooled test uses

$$N = 5 \times 1000 = 5000$$

observed deaths. Because $m = 27$ parameter settings were tested, a Bonferroni correction was applied at familywise level ($\alpha = 0.05$). The corrected threshold is

$$\alpha_{\text{Bonferroni}} = \frac{0.05}{27} \approx 0.00185185.$$

A parameter setting is therefore considered significant only if

$$p_{\text{pooled}} < 0.00185185.$$

The pooled results show a robust enrichment of observed deaths in the T-zone across all tested death/ERK parameter settings. The pooled T-zone fraction ranges from

$$0.5742 \quad \text{to} \quad 0.6642,$$

which is always substantially above the geometric null value

$$p_0 = 0.4444.$$

The pooled T-zone density ratio ranges from

$$1.686 \quad \text{to} \quad 2.472.$$

Therefore, even in the weakest tested case, the observed death density inside the T-zone is approximately 1.69 times the density outside the T-zone. In the strongest tested case, the density inside the T-zone is approximately 2.47 times the density outside the T-zone.

All 27 pooled binomial tests are significant after Bonferroni correction. The strongest pooled T-zone enrichment occurs for

$$\lambda_d = 0.5, \quad \beta_{d,R} = 4.0, \quad \beta_{d,T} = 1.6,$$

corresponding to the parameter setting

$$\text{deathERK_ld0p5_bdR4p0_bdT1p6.}$$

For this setting,

$$k_T = 3321, \quad N = 5000,$$

so that

$$\hat{\pi}_T = \frac{3321}{5000} = 0.6642,$$

with pooled density ratio

$$R_T = 2.472,$$

and one-sided pooled binomial p -value

$$p_{\text{pooled}} = 9.06 \times 10^{-215}.$$

The weakest pooled enrichment occurs for

$$\lambda_d = 2.0, \quad \beta_{d,R} = 1.0, \quad \beta_{d,T} = 0.4,$$

corresponding to the parameter setting

$$\text{deathERK_ld2p0_bdR1p0_bdT0p4.}$$

Even in this weakest case,

$$k_T = 2871, \quad N = 5000,$$

so that

$$\hat{\pi}_T = \frac{2871}{5000} = 0.5742,$$

with pooled density ratio

$$R_T = 1.686,$$

and one-sided pooled binomial p -value

$$p_{\text{pooled}} = 1.02 \times 10^{-75}.$$

This value is still far below the Bonferroni threshold (0.00185185).

The death and ERK parameters mainly affect the temporal speed of the observed death process. Averaged over all other ERK parameters, increasing λ_d decreases the mean time required to obtain 1000 analyzed observed deaths:

λ_d	mean time for 1000 deaths	mean T-zone fraction	mean density ratio
0.5	114.51	0.6370	2.218
1.0	75.20	0.6232	2.081
2.0	53.32	0.6081	1.963

Thus, larger λ_d accelerates the accumulation of observed deaths. However, the T-zone fraction remains clearly above the null area fraction ($p_0 = 0.4444$) for all tested values of λ_d .

Similarly, increasing $\beta_{d,R}$ decreases the mean ERK radius, since

$$\mathbb{E}[R^d] = \frac{1}{\beta_{d,R}},$$

and increasing $\beta_{d,T}$ decreases the mean ERK lifetime, since

$$\mathbb{E}[T^d] = \frac{1}{\beta_{d,T}}.$$

Both changes reduce the strength or duration of ERK protection and therefore tend to accelerate the observed death process. Nevertheless, the T-zone enrichment remains present throughout the full death/ERK parameter grid.

The fastest parameter setting is

$$\text{deathERK_ld2p0_bdR4p0_bdT1p6},$$

with a mean time of approximately

$$20.53$$

to obtain 1000 analyzed observed deaths. The slowest parameter setting is

$$\text{deathERK_ld0p5_bdR1p0_bdT0p4},$$

with a mean time of approximately

$$214.84.$$

This confirms that the death/ERK parameters strongly control the time scale of the observed death process.

Overall, these results show that T-zone enrichment is statistically overwhelming and robust across the tested death/ERK parameter sweep. The candidate death intensity λ_d , the ERK radius parameter $\beta_{d,R}$, and the ERK lifetime parameter $\beta_{d,T}$ modulate the speed and, to a lesser extent, the strength of the enrichment. However, they do not remove the T-zone enrichment under the fixed activation-localization regime. Therefore, the main conclusion is that death/ERK parameters primarily control the temporal dynamics, while the existence of the T-zone enrichment is robust and mainly inherited from the fixed activation localization structure.